

# होमवर्क साथी

# Homework Saathi

A Hindi homework helpline for the *parents* of Class 5 government-school children in India — built around the one thing it must never do, which is guess.

Class 5 • गणित

NCERT गणित मेला • 15 chapters • 190 pages

RAG • refusal-first

Zero paid APIs

Try it — ask it something →

Runs on a free instance that sleeps when idle — the first load can take about a minute to wake.

## The user is not the child

A parent in a government-school district opens WhatsApp at night because their Class 5 child is stuck on a maths problem. Many of these parents left school before the class their child is now in. They cannot check a maths answer — which is precisely why they are asking.

That single fact settles almost every feature argument, so it is written down as the product's first line:

हम बच्चों के ट्यूटर नहीं हैं — हम माता-पिता के कोच हैं।

We are not a tutor for children. We are a coach for parents.

So the output is not an answer. It is **a sentence the parent can say out loud to their child**, plus the reasoning that earns it, plus the page of the real textbook it came from.

साथी

किताब से

1 जवाब

एक किलोग्राम में 1000 ग्राम होते हैं।

2 एक नियम

किलोग्राम को ग्राम में बदलने के लिए 1000 से गुणा कीजिए।

3 अलग संख्याओं से उदाहरण

2 किलोग्राम को ग्राम में बदलने पर 2000 ग्राम मिलते हैं।

4 बच्चे से यह कहिए

“बेटा, एक किलो में हजार ग्राम होते हैं।”

किताब: अध्याय 8 — भार और धारिता, पेज 111

साथी

जवाब नहीं दूँगा

यह सवाल कक्षा 5 की किताब से आगे का है — यह गणित तो है, पर इस किताब में नहीं सिखाया गया। बड़ी कक्षा में यह आएगा, या शिक्षक जी से पूछ लीजिए।

पूछा गया: “द्विघात समीकरण का सूत्र क्या है?” — कक्षा 10 बीजगणित

Left: the four-part answer contract, enforced as code — an answer that fails it is refused, not shipped. Right: the same system declining Class 10 algebra. Both are real output, captured from the running app.

## The hard problem is not generating. It is refusing.

A confidently wrong maths answer that a parent then teaches their child is far worse than “*I don't know*” — because the parent has no way to detect the error. Fluency is the failure mode, not the goal.

So the system is built back-to-front from that: a refusal gate calibrated on a 150-question labelled set, and a contract validator that turns a bad generation into an honest decline rather than a plausible paragraph.

# 87%

of answerable questions get answered

n = 150 labelled

# 1.1%

wrong-answer rate — 1 of 88 answered

the number that matters

# 13%

of legitimate questions refused

guardrail: 25%

# 93%

retrieval returns the right passage

chunk-level, top-3, n = 30

# 0.7%

character error rate reading the book

100% numeric & fraction recall

# 83%

answers meeting the four-part contract

n = 30, median 6.0s

The gate and retrieval figures are reproducible with no LLM budget at all — `calibrate_refusal.py` and `eval_retrieval.py`. Every figure here was re-measured rather than carried forward from notes.

## Five times the evidence overruled me

The decision log records reversals as prominently as decisions, because the reversals are where the product was actually made. These five changed it most.

DO

### I chose OCR, and OCR was the wrong tool

Scanning the pages seemed obviously right for a textbook full of diagrams. It gave 7.9% character error — and Tesseract's Hindi model *deletes every digit 1*, reading it as a Devanagari full stop. 150 became 50. In a maths product that is not a quality issue, it is a disqualification.

~~OCR — 7.9% CER~~

→

PDF text layer + normaliser · 0.7% CER

**What it changed:** I had blamed "the text layer" generically for corruption that belonged to one specific library. Naming the real culprit turned a rejected approach into the winning one.

## D6 Parents do not use the textbook's words

Of 51 everyday Hindi terms a parent would actually say, **24 appear nowhere in the book**. नक्शा (map) occurs zero times — even though an entire chapter teaches maps. The book says मानचित्र. I was one step from labelling those chapters "not answerable".

~~relabel 2 chapters unanswerable~~ → 22-entry audited vocabulary map

**What it changed:** a retrieval miss that looks like missing content is often a vocabulary gap between the user and the corpus. The content was there the whole time.

## D1 The obvious knob was the wrong knob

The plan was to sweep a similarity threshold until refusals looked right. Similarity alone yields **1% coverage** — useless. And across the range 0.30–0.475 the curve is flat: the threshold barely matters. What does the work is a stack of categorical layers, including 516 out-of-syllabus chunks from Classes 6–9 that exist only to be matched against.

~~one-tuned threshold~~ → layered gate · 1% → 87% coverage

**What it changed:** without decoys, "is this Class 5 content?" is a question the corpus cannot answer — every chunk in it *is* Class 5. The decoys made the boundary real.

## D14 30% of the textbook was invisible, silently

Chunk IDs collided. A dictionary keyed by ID dropped **74 of 248 chunks** with no error, no warning, nothing. A third of the book was missing from retrieval — and every metric measured before that discovery had been measured over the remainder.

~~174 chunks, unknowingly~~ → 248 chunks, asserted at both ends

**What it changed:** the fix was cheap; noticing was not. Uniqueness is now asserted where IDs are written *and* where they are loaded, and neither assertion may be relaxed.

## D2 Blame your own instrument first

Contract conformance sat at 56%, and the model looked like the problem. Reading the actual failures found **five defects in my validator** — including one where my prompt's word for "passage" (अंश) also means "numerator", so the model was marked wrong for echoing my own term.

56.5% — ~~"the model is weak"~~

→

83% — the ruler was bent

**What it changed:** before tuning the thing being measured, verify the thing doing the measuring. Reading raw output beat every aggregate metric.

## How it works

1

### Query pre-checks

Deterministic, before any retrieval or model call. Catches 36 of 50 refusable questions for free — including requests to simply do the homework, which are refused on purpose.

2

### Hybrid retrieval

BGE-M3 dense cosine plus IDF term overlap across 248 concept-unit chunks. Dense alone scored 87%; the lexical signal took it to 93%.

3

### A syllabus boundary made of decoys

516 chunks from Classes 6–9. If a question sits closer to a Class 7 page than a Class 5 one, it is out of syllabus.

4

### Generation

Groq's free tier, in Hindi, against the top three passages.

5

### Contract validation

Four parts — answer, rule, a worked example using *different* numbers, and the sentence to say to the child — enforced as code. Fail the contract, and the system refuses instead of shipping it.

## Constraints, taken as product decisions

- **Zero paid APIs.** Free tiers only. That is a decision about cost-per-conversation for users who cannot pay, and a sovereignty one — Bhashini (Government of India) for voice rather than a US frontier API. The ceiling it buys is roughly **90 answers a day**, and the cost model says so out loud.
- **An 8 GB laptop.** It killed the 7B local vision model at 980 seconds per page. Local models cap around 2B, and the architecture reflects that rather than pretending otherwise.
- **The textbook is © NCERT, “not to be republished.”** So the deployed app ships no page images. When a parent asks to see the page, it fetches that chapter from NCERT's own server at request time and renders the single page. Nothing is redistributed.

## What is not done

Listed at the same weight as the results, because a case study that only reports wins is not evidence of judgement.

- **Parent discovery interviews.** The riskiest assumption in the entire thesis — that a parent accepts “I don't know” rather than leaving — is *untested*. It is the next thing to do and it gates everything.
- **Every test question is authored by me**, not observed from a real parent. The retrieval numbers are optimistic until that changes.
- **Voice** is wired to Bhashini but awaiting credentials; the browser speech API stands in and is labelled honestly where it is used.
- **Reading figures is built but switched off** — a vision model can read a bar chart off the page, but the deployment runs without that key, so picture-only questions are refused with a reason instead.
- **WhatsApp delivery** is not built. The web chat exists because the WhatsApp test number can only message pre-approved recipients — a link a stranger can open is the only demo that actually demonstrates.

The refusal rate on legitimate traffic — 13% — is inside the 25% guardrail, but it is still 13 parents in 100 told “not this time”. Whether that is a tolerable experience or a fatal one is

exactly what the interviews must answer, and no amount of further tuning substitutes for asking them.

---

## Build

Python standard-library HTTP server, no framework — a demo that needs an install step before it runs is a worse demo. BGE-M3 embeddings over a NumPy index. Groq free tier for generation. 51 Playwright assertions covering the interface, including that no internal reason code ever reaches the parent's screen. Zero npm.

The decision log is the primary artefact: every decision with its evidence, and every reversal with the number that forced it.

---

**Pradyumn Awasthi** — AI Design Associate, ConveGenius. Built as a product portfolio project.

Textbook: NCERT गणित मेला (Maths Mela), Class 5, NCF-2023 — fetched from [ncert.nic.in](https://ncert.nic.in) and never redistributed.